

Heart Disease Risk Assessment Report

Heart Disease Risk: 96.4%

Classification: HIGH RISK

Disease-Specific Risk Scores

Condition	Risk %
Coronary Artery Disease	99.1%
Hypertension	98.5%
Heart Failure	90.9%
Arrhythmia	83.8%

Patient Parameters

Parameter	Value	Description
age	58	Age (years)
sex	1	Sex (1=Male, 0=Female)
cp	0	Chest Pain Type (0-3)
trestbps	150	Resting Blood Pressure (mmHg)
chol	280	Serum Cholesterol (mg/dl)
fbs	1	Fasting Blood Sugar > 120mg/dl (1=Yes)
restecg	1	Resting ECG Results (0-2)
thalach	120	Max Heart Rate Achieved
exang	1	Exercise Induced Angina (1=Yes)
oldpeak	3.5	ST Depression (exercise vs rest)
slope	0	Slope of Peak Exercise ST Segment
ca	2	Number of Major Vessels (0-3)
thal	2	Thalassemia (0-3)
bmi	31.2	Body Mass Index
hdl	32	HDL Cholesterol (mg/dl)
ldl	195	LDL Cholesterol (mg/dl)
triglycerides	310	Triglycerides (mg/dl)
smoking	1	Smoking Status (1=Yes)
family_history	1	Family History of Heart Disease (1=Yes)
diabetes_history	1	Diabetes History (1=Yes)
hba1c	7.8	HbA1c (%)

Key Risk Drivers

- ↑ Body Mass Index: 31.2 ■ High (normal <24.9) [+42.099998474121094% risk contribution]

- ↑ Age (years): 58 [+34.0% risk contribution]
- ↑ Resting Blood Pressure (mmHg): 150 ■ High (normal <120) [+32.20000076293945% risk contribution]
- ↑ HDL Cholesterol (mg/dl): 32 ■ Low HDL [+31.899999618530273% risk contribution]
- ↑ Smoking Status (1=Yes): 1 [+31.100000381469727% risk contribution]

Health Recommendations

Diet

- Reduce saturated fats; choose olive oil, nuts, and avocados.
- Eat fatty fish (salmon, mackerel) 2× per week for Omega-3.
- Limit refined carbs and sugars; avoid sweetened beverages.
- Target a calorie deficit of 300–500 kcal/day for gradual weight loss.
- Increase HDL by consuming whole grains, beans, and moderate nuts.
- Adopt a low glycaemic-index diet; limit white rice and bread.
- Follow a DASH diet; reduce sodium intake to <2,300 mg/day.
- Eat ≥5 servings of vegetables and fruits daily.
- Stay well hydrated — aim for 8–10 glasses of water per day.

Exercise

- Start with supervised cardiac rehabilitation exercises.
- Begin with 15–20 min of light walking; increase gradually.
- Low-impact activities (swimming, elliptical) to protect joints.
- Join a smoking cessation program — smoking multiplies cardiac risk.

Medical Consultation

- ■ HIGH PRIORITY: Consult a cardiologist within the next 7 days.
- Request: Stress ECG, Echocardiogram, Full Lipid Panel, HbA1c.
- Endocrinologist consultation for diabetes management.
- Inform cardiologist of family history for genetic risk screening.

This report is generated by an AI model for informational purposes only. Always consult a qualified medical professional for diagnosis and treatment.